The Earth as an electromagnetic machine

Geodynamo, global atmospheric circuit and Schumann cavity as Kron networks — CLAUDE.md §3a, §3f

kronspice simulation report · 2026-11-07 · dimensionless Rikitake μ = 1, A = 1

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Summary

The planet is modelled as three Kron networks built from kronspice elements. Layer 1 is the core dynamo as Rikitake's two-disk dynamo (two cross-coupled homopolar machines, machine elements N). Layer 2 is the global atmospheric electric circuit: storms charge the ionosphere through leaky fair-weather air, modelled as 60 shells of R || C. Layer 3 is the Earth–ionosphere cavity as a transmission line in colatitude, which carries the Schumann resonances. The layers are then linked in one network, the dynamo is put through convergence, parameter and drive studies, a stacked two-disk design is tested, and a nonlinear (electron-heated) ionosphere is added.

- **The Rikitake dynamo reverses chaotically:** 71 reversals in t = 50…1000, intervals 3.92–70.36, mean 12.17, CV 0.92 (Poisson: 1). Equilibrium $I_1^2 = 1.618034$ (the golden ratio), natural period 4.2018.
- **Energy closes:** drive work 2581 = heat 2578 + Δstored 3.078, residual 1.8×10^{-5} %; engine ledger mismatch 1.1×10^{-5} J (4×10^{-9} of the drive work).
- **Global circuit:** calibrated on 250 kV, 2 pA/m² and 120 V/m, it gives 1020 A, 244.9 Ω, 255 MW, an Earth charge of -5.42×10^{5} C, and a relaxation time $\varepsilon_0/\sigma_0 = 531$ s. Half the balancing positive charge sits below 1.42 km.
- **Schumann cavity:** the perfect-cavity ladder equals Legendre's $c/(2\pi a)\cdot\sqrt{n(n+1)}$ to 4×10^{-4}. A two-height ionosphere fitted to the five measured modes reproduces them within 2 % (rms error 0.0095), with predicted Q_2–Q_5 of 4.7–4.2.
- **One network for layers 2 and 3:** the same elements give 249.59–250.27 kV at DC and ELF peaks 7.2, 13.0, 18.9, 24.9, 30.9 Hz with no fitted electric height (measured 7.8, 14.3, 20.8, 27.3, 33.8 Hz).
- **Stacked disks do not reverse:** any self-coupling drains the speed difference $A = \omega_1 - \omega_2$ and the field locks to one polarity; no gap from 0.53 to 6 r_o reverses.
- **No layer powers another:** every drive study (paper107 variations a–e) closes its ledger; none adds energy.

What is being modelled

This example is standard physics, not a register hypothesis. It applies Kron's network method (§3a) and the homopolar machine of §3f to the Earth. The paper107 drive studies (Evans, Amador & Eckardt) give the standard Maxwell–Lorentz counterpart of the ECE reading, which stays in the register as H15 and H16. The dynamo results are dimensionless: there is no claim about reversal times in years.

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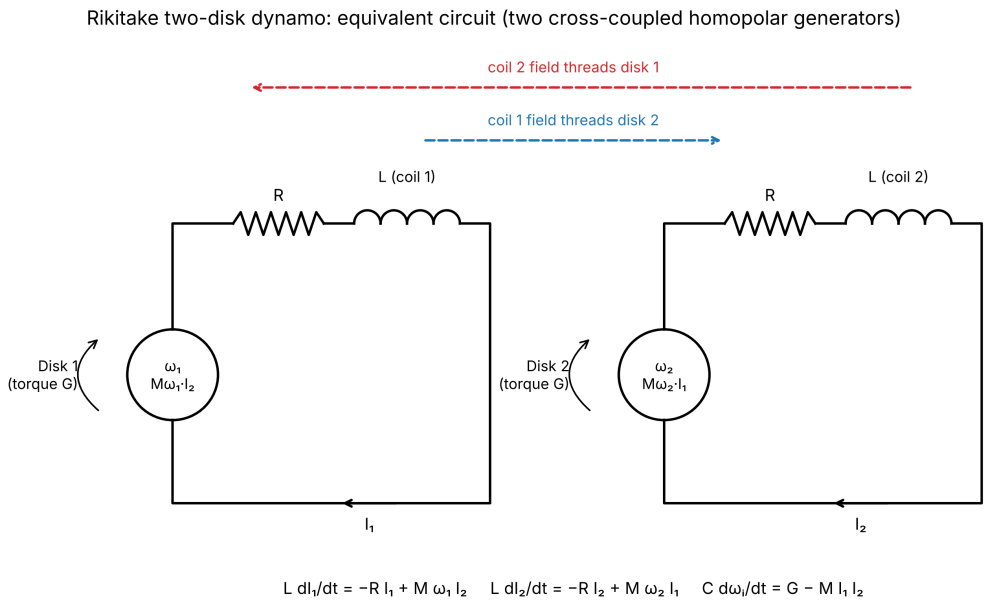


Figure 1. Equivalent circuit of the Rikitake dynamo: two loops, each with a disk (speed EMF $M\omega_i \cdot I_j$), a resistance R and a field coil L whose field threads the other disk. Each shaft is driven by a torque G.

Layer 1 — the Rikitake two-disk dynamo

Model

part	built as
disk *i*	machine element with two windings: its radial path and the other loop's field coil; speed matrix G[0][1] = ±M gives the speed EMF $M \cdot \omega_i \cdot I_{coil}$ and the back torque $M \cdot I_1 \cdot I_2$
shaft *i*	mechanical node, inertia C (capacitor), drive torque G (current source on the mechanical channel)
loop *i*	disk path → R → field coil L (threads the other disk)

The equations are $L\,dI_1/dt = -R\,I_1 + M\,\omega_1\,I_2$, $L\,dI_2/dt = -R\,I_2 + M\,\omega_2\,I_1$, $C\,d\omega_i/dt = G_i - M\,I_1\,I_2$. With all constants 1 this is Rikitake's dimensionless system with μ = 1, A = 1. The linearization about the equilibrium is $(\lambda + 2)(\lambda^2 + k^2 + k^{-2})$: a neutral oscillation of period 4.20, the floor of the reversal intervals.

Four cases

case	reversals	intervals (min / mean / max)	CV	energy residual (trace)	engine ledger
1 co-rotating (Rikitake)	**71 (RK4: 49)**	**3.92 / 12.17 / 70.36**	**0.92**	**1.8×10^{-5} %**	**1.1×10^{-5} J**
2 disk 2 reversed, same wiring	0	—	—	6.7×10^{-5} %	-4.2×10^{-5} J
3 disk 2 reversed, coil 2 rewound	71 (RK4: 49)	3.92 / 12.17 / 70.36	0.92	1.8×10^{-5} %	1.1×10^{-5} J
4 co-rotating, coil 2 rewound (Tesla's opposed fields)	0	—	—	6.7×10^{-5} %	-4.1×10^{-5} J

- Case 1: the circuit tracks an independent RK4 integration to 0.0041 for t ≤ 10; after that chaos separates them, so only the statistics are comparable. Final shaft speeds $\omega_1 = 2.47$, $\omega_2 = 1.47$.
- Case 2: the feedback is negative; $|I_1|$ after t = 50 is 6.67×10^{-14} and both shafts spin up without load (ω = ±10³).
- Case 3 equals case 1 with I_2 and ω_2 negated (test TestCounterRotatingRewoundMirrors, to 10^{-9}): the sense of the feedback loop matters, not the direction of spin.
- Case 4: about I = 0, λ = −1 ± i; the field dies. Tesla's magnets excite his disks separately.

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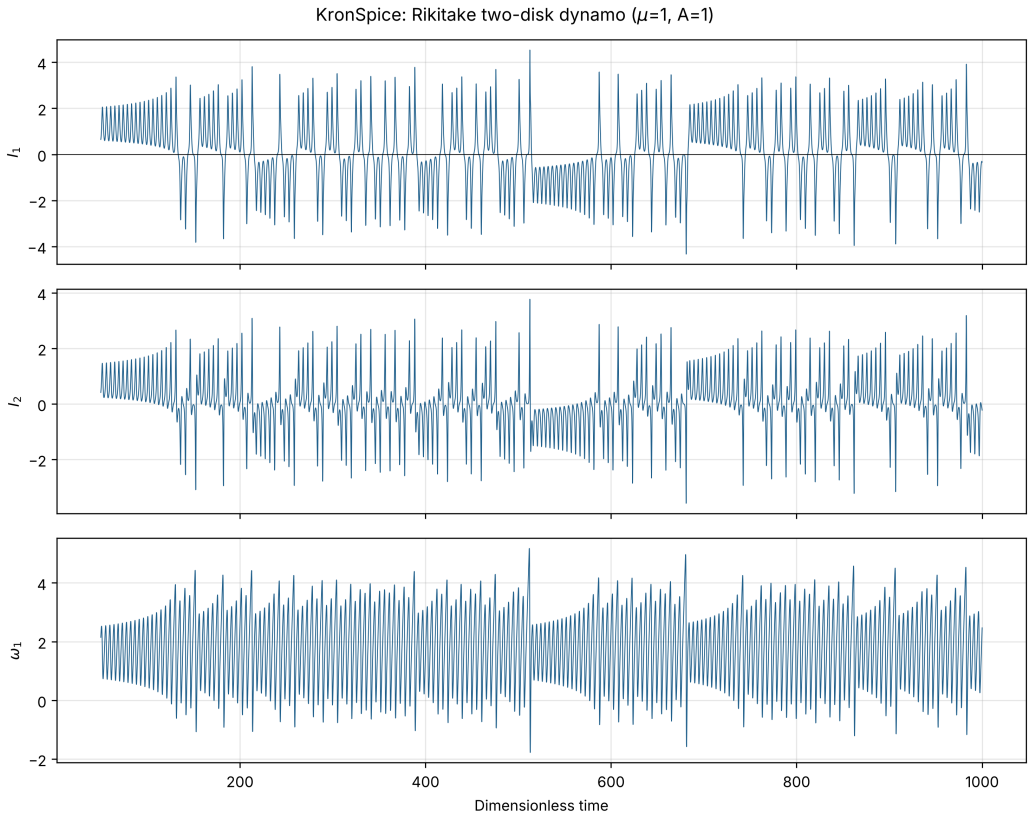


Figure 2. Case 1 transient from kronspice: I_1, I_2 and ω_1 against dimensionless time (t = 50…1000). Polarity reversals appear as sign changes of I_1.

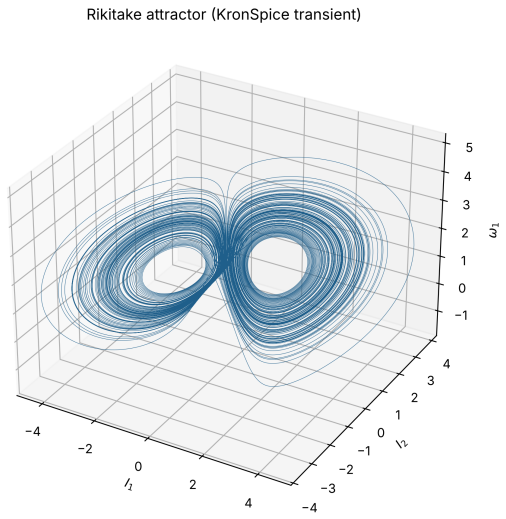


Figure 3. The Rikitake attractor traced by the kronspice transient in (I_1, I_2, ω_1).

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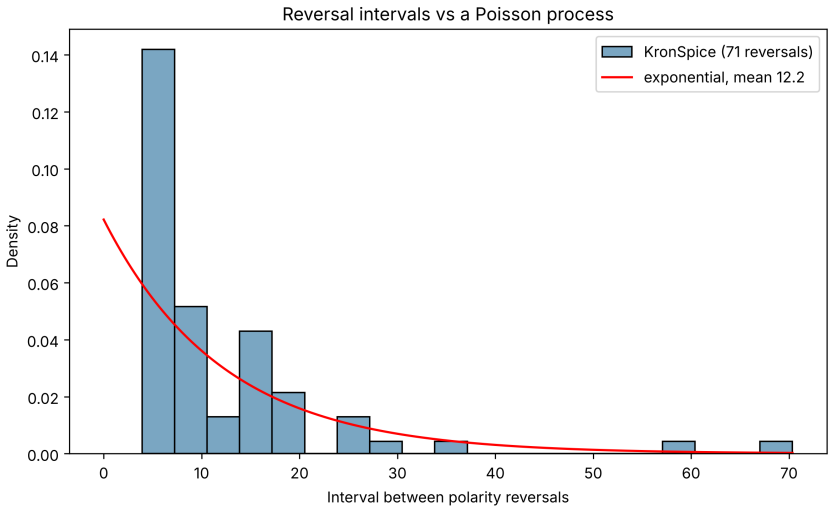


Figure 4. Histogram of the 71 reversal intervals of case 1 against an exponential (Poisson) density with the same mean, 12.2. No interval is shorter than about 4, the oscillation period.

Layer 1 studies: convergence and the (μ, A) plane

Counts at a fixed time depend on the integrator; the interval distribution does not. RK4 over t = 20 000:

step h	reversals	mean	CV	KS vs h = 0.0005: D, p
0.004	1385	14.35	0.99	0.062, 0.01
0.002	1442	13.78	0.99	0.027, 0.65
0.001	1470	13.52	1.00	0.022, 0.88
0.0005	1473	13.49	1.01	—

The kronspice circuit at maximum step 0.02, 0.01 and 0.005 (t = 4000) gives 269, 307 and 291 reversals and KS p = 0.13, 0.38, 0.39 against the finest RK4. A parametric bootstrap KS test (400 resamples) rejects exponential intervals at every (μ, A) point tried (p = 0.0025, the floor). μ = 0.5 with A ≥ 2 is periodic (CV 0.00). Longest/mean is at most 6.0, against of order 100 or more in the polarity record (the Cretaceous Normal Superchron, ≈ 38 Myr, against a Cenozoic mean of 0.2–0.3 Myr). Where the machine is chaotic (μ = 1, 2), the last swing before a reversal is 3–23 % above the median swing (ratio 1.03–1.23), whereas the record shows the dipole weakening before it reverses (Valet & Meynadier 1993).

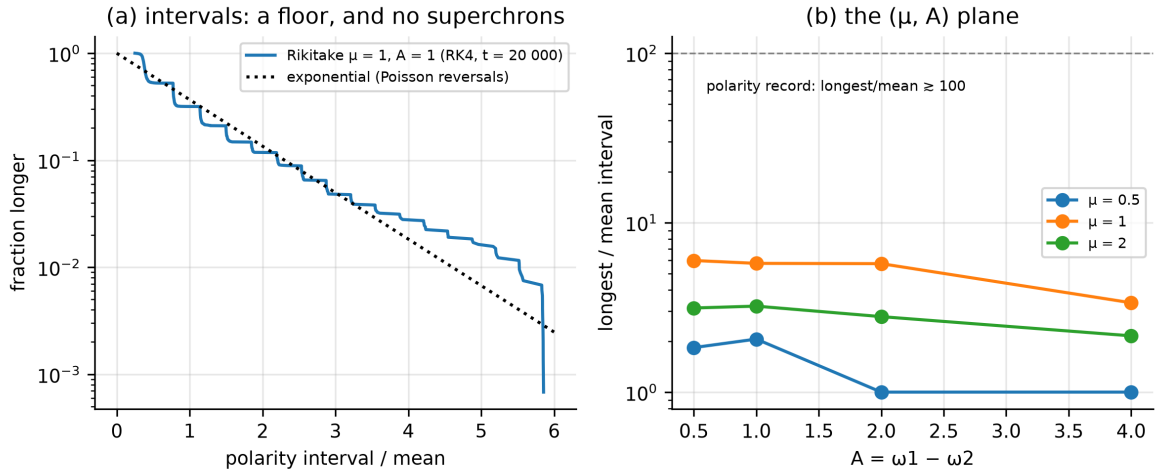


Figure 5. (a) Fraction of intervals longer than a given multiple of the mean, Rikitake μ = 1, A = 1 (RK4, t = 20 000), against an exponential: a floor and no superchrons. (b) Longest/mean interval over the (μ, A) plane; the polarity record has of order 100 or more.

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paper107 drive variations (task 4)

variation	built as	result
a. modulated torque G = 1 + 0.3 sin($\omega_d t$)	SIN current sources on the shafts	reversal rate 0.0326–0.0865 per unit time over the run_output sweep (0.0747 unmodulated); RMS I_1 1.313–1.343; circuit vs RK4 $\lvert\Delta I_1\rvert \le$ 0.0076
b. each disk on its own speed drive	voltage sources on the mechanical nodes	growth $(-R + M\sqrt{(\omega_1\omega_2)})/L$; circuit = exact to 4 decimals; opposed speed modulation decays at the old threshold
c. feedback coil around each disk	G[0][0] = −K·M	K ≥ +0.25: no reversals; K = −0.5: 21 reversals against 1438; circuit vs RK4 $\lvert\Delta I_1\rvert \le$ 0.0014
d. Tesla's opposed fields	case 4	$\lambda = -1 \pm i$ about I = 0; $\lvert I_1\rvert$ after t = 60 is 1.7×10^{-27}
e. rotating-frame terms	one disk at ω(t) = 5 sin 10t	machine = exact to 7×10^{-6}; for a 1 T, 10 cm, 50 rev/s disk the Stewart–Tolman, centrifugal and Coriolis terms are 6×10^{-11}, 2×10^{-9}, 4×10^{-9} of the 1.571 V motional EMF

None adds energy and every ledger closes. The feedback coil changes how the drive's energy splits between field and heat: its EMF K·M·ω·I is paid by the back torque K·M·I² on the shaft.

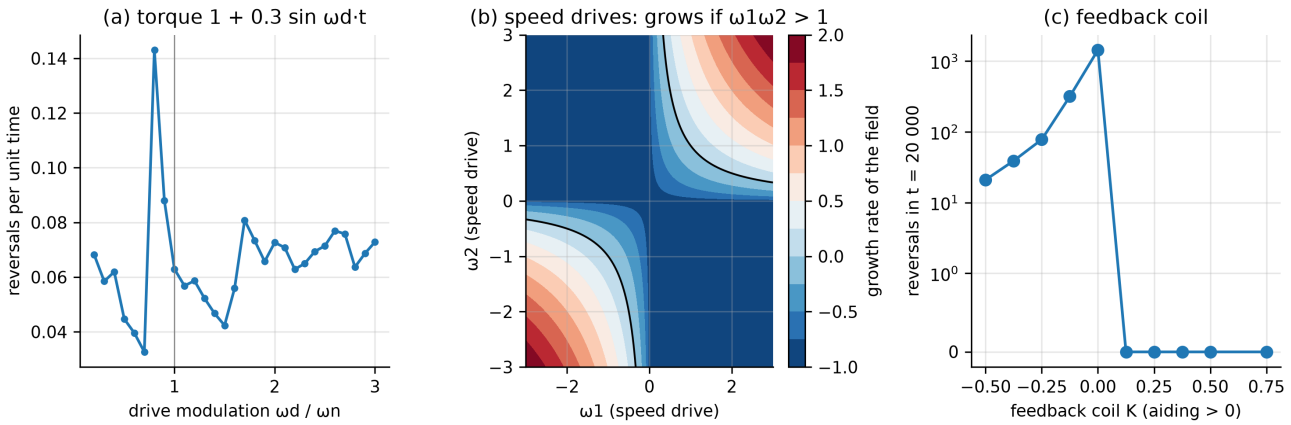


Figure 6. (a) Reversal rate against torque modulation frequency ω_d/ω_n (a finer sweep than the run_output table; peak near 0.8 ω_n). (b) Growth rate of the field with both disks on speed drives: it grows only where $\omega_1\omega_2 > 1$. (c) Reversals in t = 20 000 against the feedback-coil gain K.

Layer 2 — the global atmospheric circuit

Wilson's circuit (Rycroft, Israelsson & Price 2000): storms lift positive charge to the ionosphere, and the return current flows down through fair-weather air. Each spherical shell of air is R || C (conduction plus displacement current). 60 shells, thin near the ground, form a ladder; the storms are one current source booked on the thermal channel. The conductivity $\sigma(z) = \sigma_0 \cdot \exp(z/H)$ is calibrated on three measured anchors, V_l = 250 kV, J = 2 pA/m² and E_0 = 120 V/m, which give $\sigma_0 = 1.67\times10^{-14}$ S/m and H = 2.08 km.

quantity	result
global current J·4πR²	1020 A
global resistance	244.9 Ω
V_l, circuit vs exact ladder	249.84 kV, rel. diff 9×10^{-6} (sphericity −0.07 % from the flat column)
Earth's charge, circuit vs $-\varepsilon_0 E_0 \cdot 4\pi R^2$	-5.42×10^5 C, both
balancing positive charge	5.42×10^5 C in the air, half below 1.42 km; 1.7×10^{-6} C at the ionosphere
capacitance	geometric shell to 60 km 0.076 F; effective Q/V_l 2.17 F
relaxation time	$R\cdot C_{eff} = \varepsilon_0/\sigma_0$ = 531 s (8.9 min)
power	255 MW from the storms, all to heat; ledger residual -4×10^{-6}
switch-on	63 % of V_l in 229 s; circuit vs exact ladder within 8×10^{-5} of V_l
Carnegie curve (±15 % daily storm current)	V_l follows with amplitude ratio 0.999691, lag 4.4 min

The charge that balances the Earth's sits in the lowest kilometres of air, not on the ionosphere, so the spherical-capacitor picture understates the capacitance about 30 times.

Refinements (declared, illustrative inputs; the three global anchors still calibrate):

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- **Two-scale profile** (boundary layer 1.5 km, free-air scale height 5 km; solved H_{BL} = 1.09 km): V_I 249.67 kV, E_0 119.8 V/m, half the space charge below 0.72 km; τ = 531 s unchanged ($R \cdot C_{eff} = \varepsilon_0/\sigma_0$ for any profile).
- **Land (29 %, σ_0 1×10^{-14} S/m) and ocean (2.2×10^{-14} S/m) in parallel:** V_I 224.8 kV; land 160 A, ocean 860 A; E_0 108 V/m over both.
- **Storm generators with their own paths** (up, down, local return), with the generator current set so that 1020 A still reaches the ionosphere; the rest returns locally:

R_{up} [Ω]	R_{down} [Ω]	R_{cloud} [Ω]	I_s [A]	reaching ionosphere	P_{gen} [MW]	in storms [MW]	fair weather [MW]	ledger
50	100	1000	1423	71.7 %	573	318	255	3×10^{-7}
100	100	500	1928	52.9 %	875	620	255	5×10^{-7}
100	500	250	4468	22.8 %	3851	3596	255	1×10^{-7}

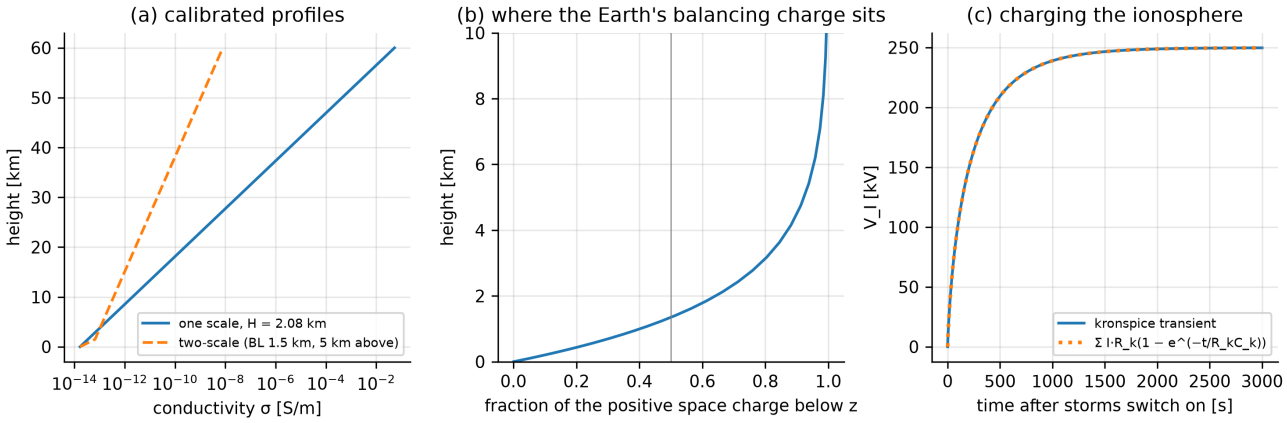


Figure 7. (a) The calibrated conductivity profiles (one scale, H = 2.08 km; two-scale). (b) Fraction of the positive space charge below height z: half lies below 1.42 km. (c) V_I after the storms switch on: kronspice transient against the exact ladder sum.

Layer 3 — the Earth–ionosphere cavity

The shell between ground and ionosphere is a line in colatitude (Madden & Thompson 1965): $V = E_r \cdot h$, $I = H_\varphi \cdot 2\pi a \cdot \sin\theta$, $L' = \mu_0 h/(2\pi a \sin\theta)$, $C' = \varepsilon_0 \cdot 2\pi a \sin\theta/h$, so the telegrapher's equations become Legendre's with $f_n = c/(2\pi a)\cdot\sqrt{(n(n+1))}$. The ladder has 181 bands of 1°; a 1 A AC current source into the pole band stands for a vertical lightning stroke.

n	perfect ladder [Hz]	Legendre [Hz]	rel.	constant height: f [Hz], Q	two-height: f [Hz], Q	measured [Hz]	two-height / measured
1	10.5910	10.5913	-3×10^{-5}	7.80, 5.0	7.96, 5.0	7.8	1.020
2	18.3432	18.3446	-8×10^{-5}	13.55, 8.7	14.27, 4.7	14.3	0.998
3	25.9393	25.9432	−0.0002	19.18, 12.2	20.64, 4.5	20.8	0.992
4	33.4841	33.4926	−0.0003	24.77, 15.8	27.11, 4.3	27.3	0.993
5	41.0044	41.0198	−0.0004	30.34, 19.4	33.68, 4.2	33.8	0.996

- **Constant heights, fitted to mode 1 only** (7.8 Hz, Q = 5; Sentman 1995): $v/c = \sqrt{(h_e/h_m)} = 0.7402$ (h_e = 49.3 km at h_m = 90 km), $\alpha = 4.93\ s^{-1}$. Modes 2–5 come out up to 10 % low (0.948, 0.922, 0.907, 0.898 of measured) and Q grows with f.
- **Two-scale-height ionosphere** (Greifinger & Greifinger 1978; Sentman 1990), with the ladder rebuilt at every frequency (781 ladders, AC ledger residual ≤ 0): h_e = 52.3 km, h_m = 90.2 km at 8 Hz, ζ_e = 0.55 km, ζ_m = 9.94 km. Q_2–Q_5 are predictions and fall inside the reported 4–6. The frequencies are well constrained; the split between the two scale heights is not.
- An equator station sees only the even modes ($P_n(0) = 0$ for odd n): peaks at 13.58, 24.78 and 35.91 Hz.

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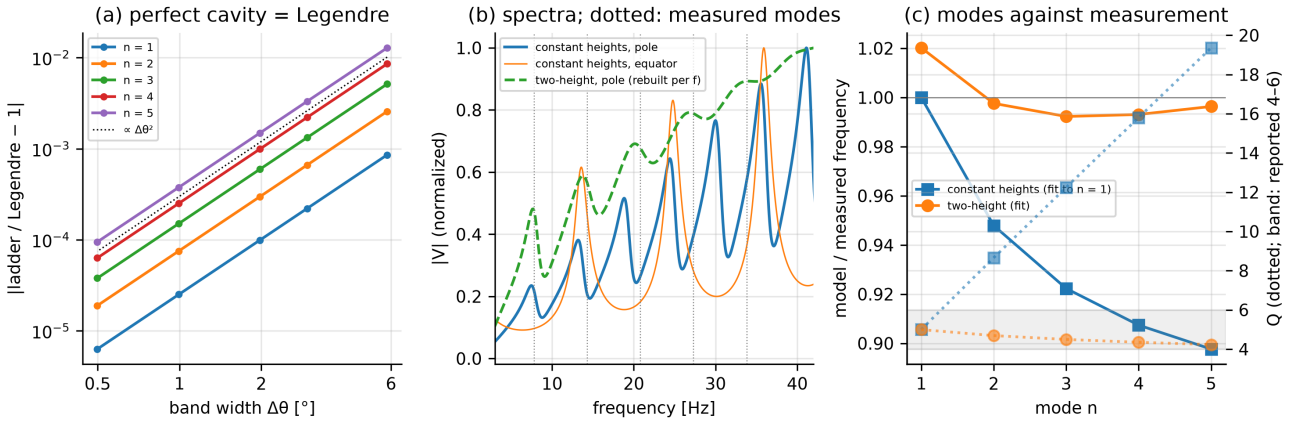


Figure 8. (a) Perfect cavity: ladder/Legendre error against band width, second order. (b) Spectra at the pole and the equator with constant heights, and at the pole with the two-height ionosphere; dotted lines are the measured modes. (c) Model/measured frequency per mode (solid) and Q (dotted, right axis; band: reported 4–6).

Linking the layers

- **One network for layers 2 and 3:** 61 bands × 40 shells to 100 km (two-scale profile), series path at h_m = 90 km. DC: ionosphere 249.59–250.27 kV across all bands, 255 MW, ledger 2×10^{-9}. ELF, same elements: peaks 7.2, 13.0, 18.9, 24.9, 30.9, 37.0 Hz at the pole and 13.5, 25.3, 37.2 Hz at the equator. The electric height comes from $\sigma(z) = \omega\varepsilon_0$ in the profile: 45.6 km at 8 Hz, 52.7 km at 33 Hz (layer-3 fit 52.3 km).
- **Dynamo → atmosphere:** the dipole shields the air from galactic cosmic rays, and σ is proportional to √q. With the unshielded ion production declared as 2× today's, V_I falls to 176.8 kV (circuit 176.7 kV) and τ to 376 s when the dipole vanishes. Over a Rikitake run V_I has a mean of 219 kV and dips to 177 kV at each reversal. The link is quasi-static: minutes for the air against millennia for a reversal.
- **One ledger:** the dynamo on the mechanical channel (geodynamo ohmic losses estimated at 0.2–0.5 TW, Christensen & Tilgner 2004), the storms on the thermal channel (0.26–3.9 GW with the storm paths); the cavity draws on lightning. No layer supplies another with energy; the dynamo only modulates the air's conductivity.

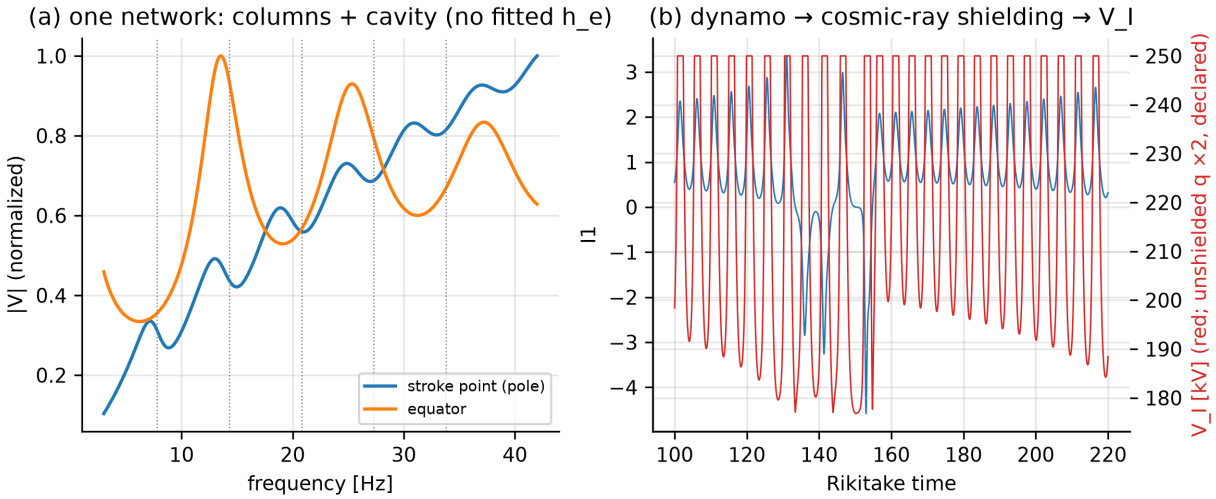


Figure 9. (a) One network of air columns plus cavity, with no fitted h_e: ELF spectrum at the stroke point (pole) and at the equator; dotted lines are measured modes. (b) Rikitake I_1 (blue) and the resulting ionospheric potential V_I (red; unshielded ion production ×2, declared).

The stacked design

The two disks (eddies) sit on one axis, each inside the coil fed by the other. The couplings come from the copper machine of examples/homopolar scaled to the disk radius (coil 1.2 r_o, 0.4 r_o long). KS is each coil's stray field on its own disk, as a fraction of M; KM is the coil–coil coupling. In the circuit the coils are inductors with a Mutual element; it equals RK4 to 0.0012 at the 0.53 r_o gap and the ledger closes (-3×10^{-6} J).

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gap [r_o]	KS	KM	reversals (t = 20 000)	RMS I_1	mean ω_1, ω_2
side by side	0	0	1438 (mean 13.76, CV 0.99)	1.343	—
0.53	0.5190	0.3870	0	0.811	0.66, 0.66
1.00	0.2630	0.1932	0	0.890	0.79, 0.79
2.00	0.0813	0.0621	0	0.962	0.92, 0.92
6.00	0.0052	0.0042	0	0.998	1.00, 0.99

Rikitake's chaos rests on a conserved speed difference $A = \omega_1 - \omega_2$: side by side, both disks feel the same back torque $M \cdot I_1 \cdot I_2$. Stacked, each disk also feels $KS \cdot M \cdot I_i^2$, which differs between the two disks, so A drains away (gap 6 r_o: A = 1.00 at t = 0, 0.77 at 50, 0.35 at 200, 0.005 at 1000) and the field locks to one polarity. Holding A with unequal drives $G = 1 \pm \delta$ brings reversals back only at gap 0.53 r_o with $\delta = 0.5$ (5119 reversals), and then like a clock: interval 3.87, CV 0.13.

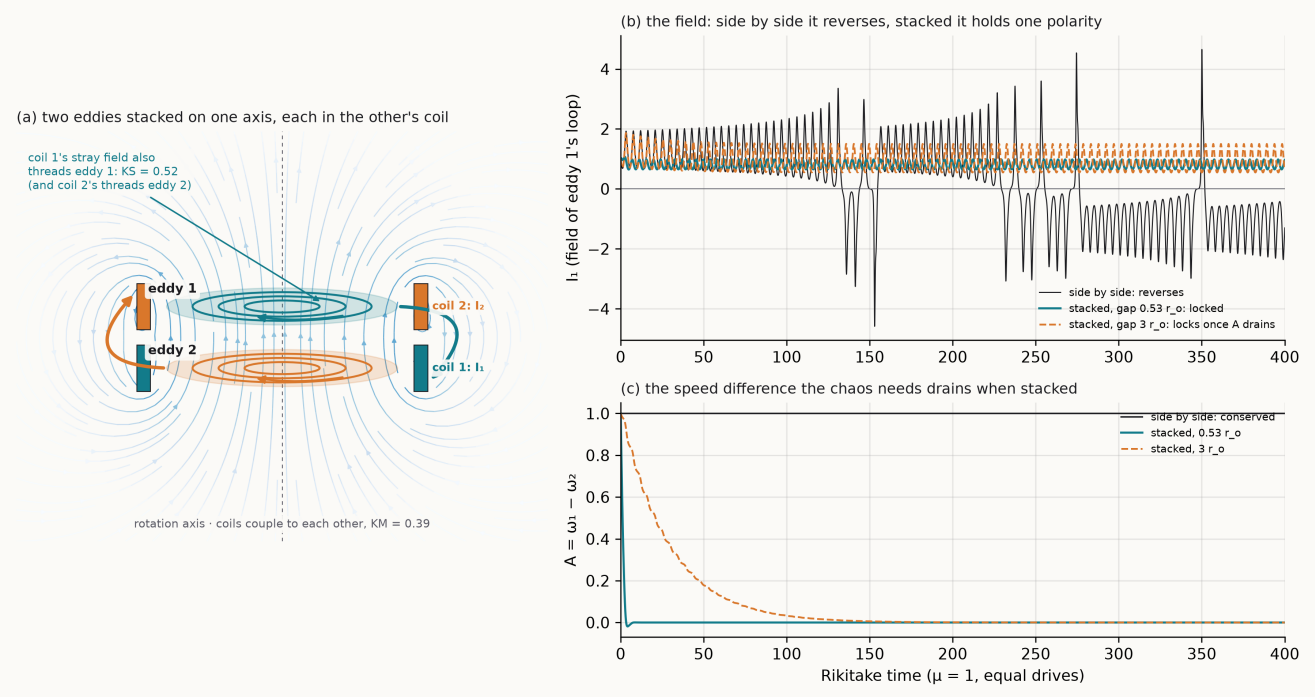


Figure 10. (a) Two eddies stacked on one axis, each in the other's coil (KS = 0.52, KM = 0.39). (b) I_1: side by side it reverses, stacked it holds one polarity. (c) The speed difference A drains when stacked.

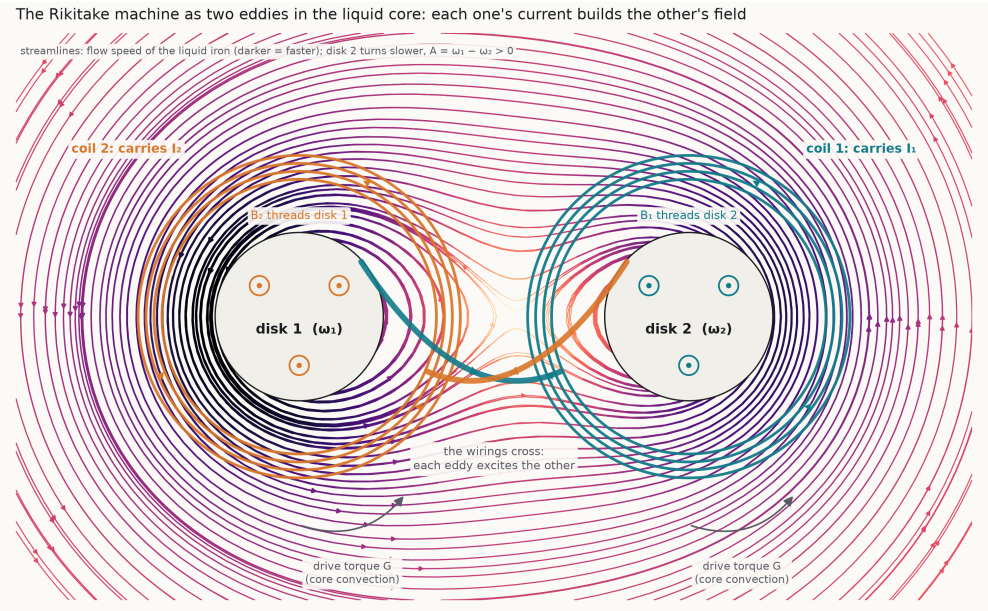


Figure 11. The side-by-side machine drawn as two eddies in the liquid core, each one's current building the field that threads the other (schematic; streamline shading shows flow speed).

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Layer 7 — a nonlinear ionosphere

The numbers in this section are from README.md; layer 7 is not part of run_output.txt. The layer-3 cavity (fitted to mode 1, Q 5) has its ionospheric loss made field-dependent by electron heating, $G(E) = G_0/(1 + (E/E_p)^2)$ (Gurevich 1978), the mechanism behind the Luxembourg effect. The heating field $E_p \approx 0.02$ V/m is declared for the night-time D region (T 200 K, δ 10^{-3}, ν 10^6 s^{-1}). The field at a band is taken as V/h_e, an upper bound, so $V_h = E_p \cdot h_e \approx 1$ kV. Two transmitters at 7.8 and 14.3 Hz drive the source band; the station is 30° away.

source peak / V_h	drive	station f_1, f_2	$2f_1-f_2$ (1.3 Hz)	$2f_2-f_1$ (20.8 Hz)	linear cavity at those lines
0.01	1 A	2.9 V	7×10^{-6} V	2×10^{-5} V	1×10^{-6}, 8×10^{-6} V
0.1	10 A	29 V	3.6 mV	3.2 mV	4×10^{-4}, 3×10^{-4} V
0.5	49 A	150 V	0.48 V	0.40 V	1×10^{-3}, 9×10^{-4} V

With heating the third-order (four-wave) products appear, and $2f_2 - f_1 = 20.8$ Hz falls on the measured third Schumann frequency. The 7.8 Hz line changes by +3 % at 0.5 V_h; its sign follows from the assumption that the loss is proportional to σ(E). The ledger closes to 3×10^{-5} in both runs. V_h corresponds to about 20 mV/m over the whole cavity height; natural Schumann fields are far smaller and real ELF transmitters stay in the linear regime, so at ELF the cavity is linear in practice.

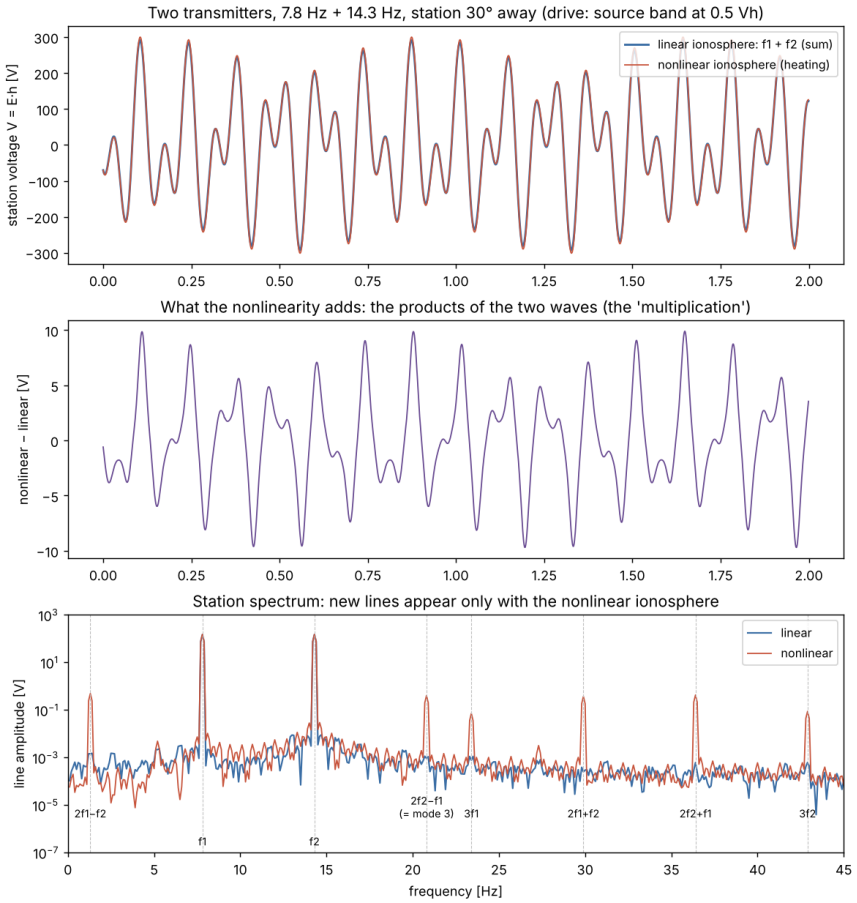


Figure 12. Two transmitters (7.8 + 14.3 Hz) at 0.5 V_h, station 30° away. Top: station voltage with a linear and a nonlinear ionosphere. Middle: their difference (the products of the two waves). Bottom: station spectrum; new lines ($2f_1-f_2$, $2f_2-f_1$, $3f_1$, …) appear only with the nonlinear ionosphere.

Ledger and validation

- Rikitake case 1: drive in 2581 = heat 2578 + Δstored 3.078 (residual 0.00047, 1.8×10^{-5} %); engine ledger 1.1×10^{-5} J. An earlier start-up mismatch of -3.75×10^4 J from the UIC point is fixed in analysis/tran.go (TestUICImposedPointNotIntegrated).
- Circuit vs RK4: case 1 0.0041 (t ≤ 10); task 4a 0.0076; task 4c 0.0014; stacked 0.0012; task 4e 7×10^{-6} of peak.

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- Global circuit: V_I circuit vs exact ladder 9×10^{-6}; switch-on 8×10^{-5}; ledger -4×10^{-6}; storm-path ledgers $1–5\times10^{-7}$.
- Cavity: perfect ladder vs Legendre $\le 4\times10^{-4}$, second order in band width; AC ledger residual ≤ 0 in every sweep; linked network DC ledger 2×10^{-9}.

Limits of the model

- The dynamo is dimensionless; there is no claim about reversal times in years.
- Rikitake gives no superchrons (longest/mean ≤ 6 against of order 100 or more) and no field weakening before reversals; self-consistent 3-D dynamos (Glatzmaier & Roberts 1995) are the benchmark it does not reach.
- Reversal counts at a fixed time are sensitive to the integrator; only interval statistics are comparable.
- The one-scale conductivity profile is an approximation; storm generator paths, land/ocean values and the unshielded ion production are declared, illustrative inputs.
- The two-height fit trades ζ_e against h_{e0}; ζ_e = 0.55 km is small for a lower-ionosphere scale height.
- Layer 7 takes the cavity's vertical field as an upper bound on what reaches the plasma, and the sign of the line shift rests on the loss being proportional to σ(E).

Reproducing the results

```
go run ./examples/earth_machine                # everything (several minutes)
go run ./examples/earth_machine -layer N       # 1 geodynamo, 2 global circuit, 3 cavity,
                                               # 4 layer-1 studies, 5 the links, 6 stacked,
                                               # 7 nonlinear ionosphere
go run ./examples/earth_machine -data viz/data  # figure data (CSV), then: python viz/plot_layers.py
go test ./examples/earth_machine
python examples/earth_machine/viz/make_report.py examples/earth_machine/earth_machine_report.pdf
```


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